SHEETH UGANTI

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SUMMARY

Second-year B.Tech AI/ML student (CGPA 9.0/10) who builds and rigorously evaluates ML systems: a GPT-2 inference engine with int8 quantization, gradient-boosting benchmarks with SHAP explainability, and a leakage-aware CERN TrackML pipeline with 220 passing tests. Generative AI paper accepted at conference proceedings. Seeking a 2027 AI/ML research internship.

EDUCATION

ATLAS SkillTech University, Mumbai 2025–2029 (expected)

B.Tech, Computer Science and Engineering (Artificial Intelligence and Machine Learning) | **CGPA 9.0/10**

- Overall course topper, Machine Learning Foundations
- Coursework: Data Structures and Algorithms, Linear Algebra, Statistics and Probability, Statistical Modelling, Calculus I–II, Mathematics for Computer Science, Computer Organisation and Architecture, Databases/SQL, NoSQL

RESEARCH EXPERIENCE

Team Lead, Harmonised Data Crosswalk and Modelling Framework (HDCMF) 2026–present

Under a university professor | Manuscript in preparation

- Unified 5 fragmented Indian household-energy sources (MoSPI HCES, Census, IRES, utility surveys, IMD climate normals) into reproducible modelling datasets by leading a team of 3 through a Python data-harmonisation workflow.
- Benchmarked 3 gradient-boosting models (XGBoost, LightGBM, CatBoost) with Spearman feature filtering on a physics-informed synthetic dataset calibrated to national statistics and BEE climate zones; XGBoost ranked best on R^2 and RMSE across both datasets.
- Identified air-conditioning ownership, cooling degree days and structure age as the leading energy-use drivers using SHAP explainability.
- Developing an application layer around the harmonised framework to support the research workflow (in progress).

Lead Researcher, GenAI for Humanity: The Future of the Next Digital Ecosystem 2025

Paper accepted, Generative AI in the Built Environment conference proceedings

- Developed the Digital Belonging Framework with 2 new measures (Participation Index, Capability Gain Metric) to assess digital inclusion and human agency, building on a systematic literature review of generative AI, adaptive personalisation and ethical AI governance.
- Paper accepted for the proceedings (Nottingham Trent University, London); delivered a 15-minute research presentation virtually.

PROJECTS

GPT-2 Inference Engine and Int8 Quantization 2026

PyTorch, Hugging Face Transformers, pytest, GitHub Actions | github.com/Sheethaleenuganti/gpt2-quant-lab

- Built a GPT-2 (124M) inference engine from scratch (attention, LayerNorm, GELU, KV cache, int8 quantizers) that matches Hugging Face logits, backed by automated tests and CI.
- Cut weight memory 51% (498 MB to 243 MB) with per-channel int8 quantization at +0.02 perplexity on WikiText-2 (25.19 vs 25.17 FP32), versus +3.52 for per-tensor.
- Diagnosed W8A8 accuracy loss (+10.58 perplexity) through activation-outlier profiling (global activation max 62.6 vs 8.6 layer mean) and per-block sensitivity analysis.
- Delivered a 4.2x to 14.9x CPU generation speedup using a KV cache across 128 to 1024 token contexts.

Particle Track Reconstruction, CERN TrackML 2026–present (ongoing)

Python, NumPy, pandas, SciPy, scikit-learn, pytest

- Built a leakage-aware reconstruction pipeline (data validation, geometric candidate-graph generation, track assembly, evaluation) with 220 passing tests and frozen, reproducible experiment artifacts.
- Audited the candidate graph on 60 development events: it recovered 503,885 of 5.0M true edges (10.06% candidate-edge recall) from 126.65M directed candidates.
- Traced the largest recall loss to the per-bin top-20 query (2.30M missed edges) plus 1.63M edges outside searched buckets, then designed one controlled candidate-generation experiment with the downstream assembler held fixed.
- Kept validation and final-test events untouched to prevent leakage.

Mini-Redis (Python, asyncio) In progress

github.com/Sheethaleenuganti/MiniRedis

- Building a Redis-compatible server with a streaming RESP2 parser, lazy and active TTL expiry, and AOF persistence with torn-write recovery; verified against redis-py with 73 tests including crash-recovery tests.
- Measured ~12,700 GET and ~9,500 SET ops/sec and quantified that always-fsync cuts write throughput about 7x.

Academic OS, Personal Academic Planning Application 2026

React, TypeScript, Firebase/Firestore, Google OAuth, Google Calendar API, Gemini API

- Designed, built and deployed a full-stack prototype integrating Google OAuth, Firestore storage, Google Calendar sync and Gemini API study guidance.

VIRTUAL PROGRAMS AND SIMULATIONS

J.P. Morgan, Quantitative Research Job Simulation Jan 2026

- Built a time-series price model on 48 months of natural-gas data capturing trend and seasonality to extrapolate prices one year forward.
- Built a prototype commodity storage-contract pricing function incorporating injection and withdrawal dates, capacity limits and storage costs.
- Built credit-risk models estimating probability of default and a FICO score rating map by bucketing scores against observed default risk.

Deloitte Data Analytics Job Simulation Oct 2026

- Designed interactive Tableau dashboards to turn complex datasets into stakeholder-ready visuals; used Excel to classify data and synthesize quantitative conclusions.

Tata iQ GenAI-Powered Data Analytics Job Simulation

- Ran GenAI-assisted exploratory data analysis to assess data quality and risk indicators, proposed a no-code delinquency-risk modelling framework, and designed an AI-driven collections strategy with ethical AI and regulatory-compliance principles.

LEADERSHIP AND INTERNATIONAL EXPERIENCE

Vice Chairman, IET Atlas Student Chapter Apr 2026–present

- Grew chapter membership by 66 through outreach and engagement; hosted a 24-hour intercollegiate hackathon with 23 teams and 5 judging panels.

Student Lead, Kindness Unlimited Nov 2025–present

- Led 55 student volunteers across community-service drives, coordinating scheduling, logistics and reporting.

Team Leader, Smart India Hackathon 2025 (Vision_X2) 2025

- Led a student team on an AI-assisted early-stage dementia detection concept; cleared the first university-level round.
- Also: Coordinator, Student Council (2025–present); Publicity Head, IEEE Women in Engineering (Oct 2025–Jul 2026); COIL Project, University of Sussex (2026), a cross-country team examining ethical use of AI in the workplace.

TECHNICAL SKILLS

Languages: Python, SQL, Java, TypeScript

Machine Learning and AI: PyTorch, scikit-learn, XGBoost, LightGBM, CatBoost, Hugging Face Transformers, LLM inference, quantization, SHAP explainability, time-series forecasting, clustering, feature engineering, model evaluation, Gemini API

Data: NumPy, pandas, SciPy, data harmonisation, data pipelines, statistical analysis, Tableau, Power BI, Excel

Tools and Engineering: Git/GitHub, GitHub Actions (CI), pytest, Jupyter, Google Colab, React, Firebase, REST APIs; learning Docker, RAG, embeddings, LangChain

Research: systematic literature review, experimental design, reproducible research, technical writing

CERTIFICATIONS

- Anthropic: Introduction to Model Context Protocol (Oct 2026); Introduction to Agent Skills (Sep 2026); AI Fluency: AI Capabilities and Limitations (Aug 2026)
- Oxford University: AI, Justice, and the Rule of Law (Oct 2026); Columbia University: AI Cybersecurity and Governance
- NVIDIA Deep Learning Institute: Generative AI; Databricks: AI Foundations; IBM: Vibe Coding (Jan 2026)
- Microsoft: Power BI (four courses); Qualcomm: AI Optimization (2026)

LANGUAGES

English, Hindi, Telugu (native); Spanish, Tamil (basic)